

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (16 pts) Let $P(-2,1)$ and $Q(4,3)$ be points in the plane.

- A. Plot P and Q on a Cartesian coordinate system
- B. Find the distance between P and Q
- C. Find the coordinates of the midpoint M of the line segment $\overline{PQ}$
- D. Find the equation of the line passing through P and Q . Write the equation in standard form.

2. (8 pts) Solve the equation $|3 - 5y| = 2$.

3. (8pts) Solve algebraically the system of equations $\begin{cases} 3x - 4y = -6 \\ 2x + 8y = 4 \end{cases}$

4. (12 pts) Let $f(x) = \frac{x^2 - 2x}{x^2 + x}$.

- A. Find the domain of f
- B. Compute the following functional values or state that the corresponding x -value is not in the domain of f

B1. $f(-2)$

B2. $f(-1)$

B3. $f(1)$

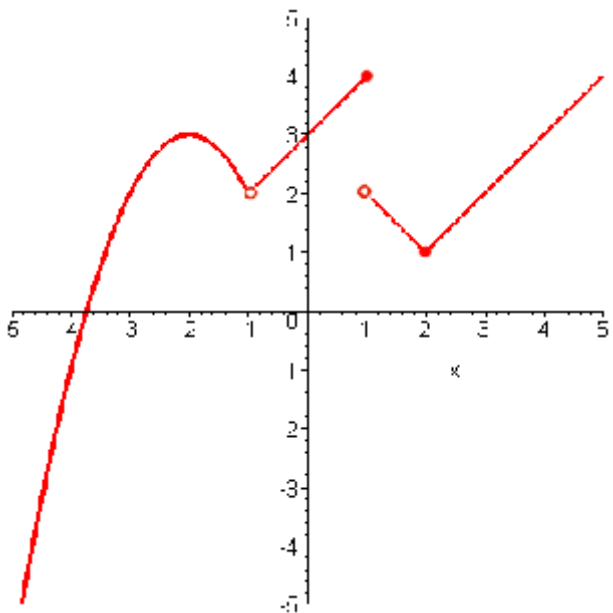
5. (12 pts) Let $f(x) = 3 + 4x$.

- A. Find the inverse of f
- B. Sketch the graphs of f and f^{-1} on same coordinate system.

6. (6pts) Find the exact value of $\sin(\cos^{-1} \frac{1}{2})$.

7. (6pts) Simplify $\sin(\tan^{-1} x)$

8. (12pts) Consider the function f defined by the graph to the right. Find each of the following (if they exist). If they do not exist, state so.



A1. $f(-1)$

A2. $f(1)$

A3. $f(2)$

B1. $\lim_{x \rightarrow -1^-} f(x)$

B2. $\lim_{x \rightarrow 1^-} f(x)$

B3. $\lim_{x \rightarrow 2^-} f(x)$

C1. $\lim_{x \rightarrow -1^+} f(x)$

C2. $\lim_{x \rightarrow 1^+} f(x)$

C3. $\lim_{x \rightarrow 2^+} f(x)$

D1. $\lim_{x \rightarrow -1} f(x)$

D2. $\lim_{x \rightarrow 1} f(x)$

D3. $\lim_{x \rightarrow 2} f(x)$

9. (24 pts) Algebraically evaluate each of the following limits.

A. $\lim_{x \rightarrow 2} x^2 - 2x - 7$

B. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$

C. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x \cos x}$